



Semester: July 2025 –November 2025		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 54 Programme: Honors in Data Science & Analytics	Class: LY	Semester: VII(SVU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: COMP	
Course Code: 116h54C701	Name of the Course: Advanced Machine Learning	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Define Machine Learning. Explain with examples how it differs from traditional programming approaches.	5
ii)	Write a note on n-grams and their importance in language modelling.	5
iii)	What is feature engineering? Explain the importance of feature selection, feature scaling, and feature extraction with examples.	5
iv)	Explain the concept of pooling. Differentiate between max pooling and average pooling.	5
v)	Write a note on evaluation metrics used in image and text classification tasks.	5
vi)	What is Transfer Learning? Explain how pre-trained CNN models can be reused for new image classification tasks	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	A dataset contains features with widely varying scales (e.g., "age" 0–100 and "income" 10,000–10,00,000). Show mathematically how Min–Max normalization or Z-score normalization can improve gradient descent performance.	10
	OR	
Q2 A	A neuron receives three inputs $x_1=0.5, x_2=0.3, x_3=0.8$ with weights $w_1=0.4, w_2=-0.6, w_3=0.2$ and bias $b=0.1$. Compute the neuron output using Sigmoid activation. What will happen if you replace the Sigmoid with ReLU?	10
Q 2 B	Solve any One	10
i)	You are training a deep network on unnormalized data. After 50 epochs, weights explode to large values and loss becomes NaN. a) Explain mathematically why this happens. b) Propose a Batch normalization technique and show its mathematical steps. c) Discuss how normalization affects the internal covariate shift.	10
ii)	a) Compare deep and shallow neural network architectures in terms of feature representation, computational cost, and learning capacity.	05
	b) Explain the concept of a Deep Multi-Layer Perceptron (MLP) with a neat	05

	diagram and mathematical representation.	
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Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Explain the step-by-step working of a 2D convolution operation with an example. Given a 5×5 input matrix and a 3×3 kernel with stride = 1 and no padding, compute the output feature map manually.	10
ii)	Explain mathematically why the vanishing gradient problem occurs in RNNs using the repeated multiplication of derivatives. Suggest and describe any two solutions.	10
iii)	Explain step-by-step how forward propagation works in an LSTM cell for a single time step. Using example inputs and arbitrary weights, compute the forget gate and input gate activations for one step (use sigmoid approximations if needed).	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Discuss the basic steps involved in building an NLP pipeline from raw text to numerical features. Explain the roles of tokenization, stemming, and lemmatization with examples.	10
ii)	Explain the Word2Vec architecture in detail. Differentiate between the CBOW and Skip-gram models. Describe how negative sampling improves Word2Vec training efficiency.	10
iii)	Given the sentences: • "The cat sits on the mat." • "The dog lies on the rug." Create Bag of Bigrams and TF-IDF representations for these two sentences. Analyze which representation better captures word co-occurrence and contextual meaning.	10

Que. No.	Question	Max. Marks
Q5	Solve any four (Short notes / Short question type)	20
i)	Write a short note on Churn Analysis. What is its importance in business decision-making?	5
ii)	Write a short note on Sentiment Analysis. How can it be applied to analyze public opinion in news articles?	5
iii)	Define Customer Segmentation. How can clustering algorithms like K-Means help identify customer groups?	5
iv)	Compare Content-Based Filtering and Collaborative Filtering approaches in recommendation systems.	5
v)	Compare Batch Gradient Descent, Stochastic Gradient Descent (SGD), and Mini-Batch Gradient Descent in terms of convergence speed and computational efficiency.	5
vi)	Derive the update rule for the Gradient Descent algorithm for minimizing a cost function $J(\theta)$.	5